Name_____ Date_____ Period_____

LESSON 4-5 · SESSION 1 · GUIDED NOTES

Find the Whole

Learning target – I can find the whole when I know a part and the percent it represents. · 6.AT.4

WORDS TO KNOW

fill in the missing words

WORD	WHAT IT MEANS	EXAMPLE
part *parte*	The amount you _____; it matches a percent.	24 stickers = 40%
whole *entero*	The full amount. It is _____.	the whole sticker pack
double number line *doble recta numérica*	Two lines that _____ amounts with percents.	$6 at 10%, $60 at 100%

KEY IDEA

complete each sentence

- The whole is 100%. When the percent is less than 100%, the whole is _____ than the part.
- Divide to find a small percent like ____, then multiply up to 100%.
- Check: the percent of your whole gives back the _____.

WORKED EXAMPLE

fill in the blanks as we solve

24 stickers are 40% of a pack. How many stickers are in the whole pack?



Step 1 40% ÷ 4 = 10%, so 24 ÷ 4 = ___ stickers.

Step 2 10% × 10 = 100%, so 6 × 10 = ___ stickers.

Step 3 Check: 0.40 × 60 = ___

Answer **The pack has ___ stickers.**

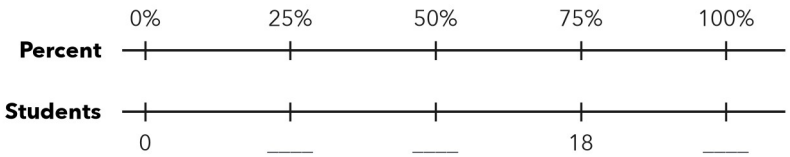
YOU TRY

use the example as a model

1. Find each whole.

a. 8 is 10% of _____ **b.** 15 is 25% of _____ **c.** 30 is 60% of _____

2. 18 students ride the bus. That is 75% of the class. How many students are in the class?



Answer_____ students

SUM IT UP

in your own words

To find the whole, I first find ___ because ___.

_____

_____

Name_____ Date_____ Period_____

LESSON 4-5 · SESSION 2 · GUIDED NOTES

Solve for the Whole

Learning target – I can write and solve an equation to find the whole from a part and a percent. · 6.AT.4

WORDS TO KNOW

fill in the missing words

WORD	WHAT IT MEANS	EXAMPLE
variable *variable*	A letter that stands for an _____ number.	w = the whole
equation *ecuación*	A math sentence with an _____ sign.	$0.6 \times w = 138$
inverse operation *operación inversa*	An operation that _____ another. Division undoes multiplication.	$w = 138 \div 0.6$

KEY IDEA

complete each sentence

- Write the equation percent × whole = part, with the percent as a _____.
- Divide the part by the decimal to find ____.
- Check: multiply your w by the percent to get the _____.

WORKED EXAMPLE

fill in the blanks as we solve

A player scored 42 points. That is 35% of the team's points. Find the team's total w.



Step 1 35% = _____

Step 2 Equation: $0.35 \times w =$ ___

Step 3 $w = 42 \div 0.35 =$ ____

Answer **The team scored ____ points.**

YOU TRY

use the example as a model

1. Solve each equation for w.

a. $0.5 \times w = 23$ _____ **b.** $0.2 \times w = 9$ _____ **c.** $0.8 \times w = 40$ _____

2. A tip of $7.20 is 15% of a bill. Find the bill.

a. Equation: _____

b. Bill: $ _____

Show your work.

SUM IT UP

in your own words

To solve for the whole, I divide because ___.

Name_____ Date_____ Period_____

LESSON 4-5 · GUIDED NOTES · MODEL, REASON, EXPLAIN

Model It. Explain It.

SESSION 1 · FIND THE WHOLE

draw it · solve it · explain it

A bike club has saved $27 for a trip. That is 30% of the money it needs. How much money does the club need in all?



a. How much money is 10%?

b. How much money does the club need in all?

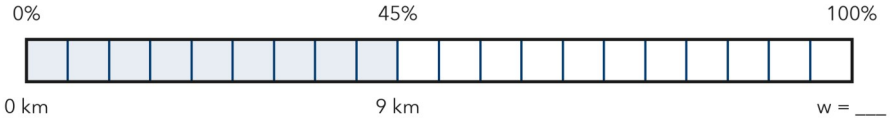
c. Explain how you know.

Challenge *Sam says the club needs 0.3 × 27 = $8.10. Show why that is too small.*

SESSION 2 · SOLVE FOR THE WHOLE

draw it · solve it · explain it

A runner has run 9 km. That is 45% of the whole race. Write an equation with w for the whole race. Solve it.



a. Write an equation for the race.

b. Solve for w.

c. Explain how you know your answer makes sense.

Challenge *Lin wrote 45 × w = 9. Show why her w is far too small. Then fix her equation.*

Check: Does my model match my numbers? Did I label units? Does my answer make sense?